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Statistics and Probability Theory – PYQ Solution (Step by Step)

Q.1 Attempt all questions

I. If two regression coefficients are -0.1 and -0.9, find r

We know:

$$r = \pm \sqrt{b_{xy} \times b_{yx}}$$

Given:

$$b_{xy} = -0.1, \quad b_{yx} = -0.9$$

$$r = -\sqrt{(-0.1)(-0.9)}$$

$$r = -\sqrt{0.09}$$

$$r = -0.3$$

Answer:

$r = -0.3$

II. Critical value for two tailed test at 5% significance level

For standard normal distribution:

$$\alpha = 0.05$$

For two tails:

$$\alpha/2 = 0.025$$

Critical value:

$$Z_{0.025} = 1.96$$

Answer:

± 1.96

III. Purpose of Null Hypothesis

Null hypothesis is used to test whether there is **no significant difference or relationship** between variables.

Answer:

It assumes that there is **no effect or no difference** in the population.

IV. Lines of regression always pass through which point?

Regression lines always pass through:

$$(\bar{x}, \bar{y})$$

Answer:

$(\bar{x}, \bar{y})$

V. Assumption required for large sample test for single mean

For large sample test:

$$n > 30$$

and population should be approximately normal.

Answer:

Sample size should be large ($n > 30$).

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Q.2 Poisson Distribution

Find probability that Ace of Spades appears at least once in 104 trials.

Probability of success in one trial:

$$p = \frac{1}{52}$$

Number of trials:

$$n = 104$$

Mean:

$$m = np = 104 \times \frac{1}{52} = 2$$

Using Poisson distribution:

$$P(X = x) = \frac{e^{-m} m^x}{x!}$$

Probability of at least one success:

$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \frac{e^{-2} 2^0}{0!}$$

$$= 1 - e^{-2}$$

$$= 1 - 0.1353$$

$$= 0.8647$$

Answer:

0.8647

Q.3 Regression Lines

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Q.3 Regression Lines

Given:

$$n = 18$$

$$\sum x^2 = 60, \quad \sum y^2 = 96$$

$$\sum x = 12, \quad \sum y = 18$$

$$\sum xy = 48$$

Step 1: Means

$$\bar{x} = \frac{12}{18} = 0.667$$

$$\bar{y} = \frac{18}{18} = 1$$

Step 2: Calculate corrected sums

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$= 60 - \frac{12^2}{18}$$

$$= 60 - 8 = 52$$

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$$

$$= 96 - \frac{18^2}{18}$$

$$= 96 - 18 = 78$$

$$S_{xy} = \sum xy - \frac{\sum x \times \sum y}{n}$$

$$= 48 - \frac{12 \times 18}{18}$$

$$= 48 - 12 = 36$$

Step 3: Regression coefficients

$$b_{yx} = \frac{S_{xy}}{S_{xx}} = \frac{36}{52} = \frac{9}{13}$$

$$b_{xy} = \frac{S_{xy}}{S_{yy}} = \frac{36}{78} = \frac{6}{13}$$

Step 4: Regression equations

Regression of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 1 = \frac{9}{13} \left(x - \frac{2}{3} \right)$$

$$13y - 13 = 9x - 6$$

$$13y = 9x + 7$$

Equation:

$13y = 9x + 7$

Regression of x on y

$$x - \frac{2}{3} = \frac{6}{13}(y - 1)$$

$$13x - \frac{26}{3} = 6y - 6$$

$$39x - 26 = 18y - 18$$

$$39x = 18y + 8$$

Equation:

$39x = 18y + 8$

Q.4 Fit Second Degree Parabola

Data:

x	0	1	2
y	1	1.8	1.3

Parabola:

$$y = a + bx + cx^2$$

Normal equations:

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

Calculate values

$$\sum x = 10$$

$$\sum x^2 = 30$$

$$\sum x^3 = 100$$

$$\sum x^4 = 354$$

$$\sum y = 12.9$$

$$\sum xy = 35.1$$

$$\sum x^2y = 123.3$$

Equations

$$12.9 = 5a + 10b + 30c$$

$$35.1 = 10a + 30b + 100c$$

$$123.3 = 30a + 100b + 354c$$

Solving:

$$a = 1.31, \quad b = -0.79, \quad c = 0.52$$

Required parabola

$$y = 1.31 - 0.79x + 0.52x^2$$

Q.5 Hypothesis Testing

Q.5 Hypothesis Testing (Mean)

Given:

$$n = 50$$

$$\bar{x} = 180$$

$$\mu = 175$$

$$\sigma = 5$$

Level of significance:

$$\alpha = 0.05$$

Step 1: Hypotheses

$$H_0 : \mu = 175$$

$$H_1 : \mu \neq 175$$

(two tailed test)

Step 2: Test statistic

$$\begin{aligned} Z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{180 - 175}{5 / \sqrt{50}} \\ &= \frac{5}{0.707} \\ &= 7.07 \end{aligned}$$

Step 3: Critical value

$$Z_{0.025} = 1.96$$

Step 4: Decision

$$7.07 > 1.96$$

Reject H_0

Conclusion

Q.6 Test of Proportion

Given:

Claim:

$$p = 0.80$$

Sample size:

$$n = 400$$

Satisfied customers:

$$x = 320$$

Sample proportion:

$$\hat{p} = \frac{320}{400} = 0.80$$

Step 1: Hypotheses

$$H_0 : p = 0.80$$

$$H_1 : p \neq 0.80$$

Step 2: Test statistic

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

where

$$q = 1 - p = 0.20$$

$$\begin{aligned} Z &= \frac{0.80 - 0.80}{\sqrt{\frac{0.8 \times 0.2}{400}}} \\ &= \frac{0}{0.02} \\ &= 0 \end{aligned}$$

Step 3: Critical value

$$Z_{0.025} = 1.96$$

Step 4: Decision

$$0 < 1.96$$

Accept H_0

Conclusion

Company's claim that 80% customers are satisfied is accepted.

Q.7 Large Sample Test

Given:

$$n = 400$$

$$\bar{x} = 10$$

$$\mu = 10.2$$

$$\sigma = 2.25$$

Step 1: Hypotheses

$$H_0 : \mu = 10.2$$

$$H_1 : \mu \neq 10.2$$

Step 2: Test statistic

$$\begin{aligned} Z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{10 - 10.2}{2.25 / \sqrt{400}} \\ &= \frac{-0.2}{2.25 / 20} \\ &= \frac{-0.2}{0.1125} \\ &= -1.78 \end{aligned}$$

Step 3: Critical value

$$\pm 1.96$$

Step 4: Decision

$$| -1.78 | < 1.96$$

Accept H_0

